

AI EDUCATION PROGRAM

AI for High School Students

A structured, hands-on curriculum taking students from foundational AI literacy to building their own subject-specific AI study agents.

Program Timeline

Goal: Build AI literacy, prompting skill, and the ability to design and build personal AI study agents — with a responsible-use mindset throughout.

1

Understanding AI (Foundations)

Demystify AI — what it is, what it isn't, and how it works conceptually.

- What is Artificial Intelligence? Clearing up hype vs. reality
- Difference between AI, Machine Learning, and Generative AI
- How large language models "think" — a non-technical explanation
- Common AI tools students already know, and how they differ
- Myths and misconceptions students commonly hold about AI

2

AI in Everyday Studies

Show practical, immediate ways AI can support learning — without replacing effort.

- Using AI for research and summarization
- Using AI to simplify difficult concepts across subjects
- Using AI for revision, self-testing, and practice questions
- Time management and study planning with AI assistance

Ethics Checkpoint: Study aid vs. academic dishonesty — discussion + examples

3

Prompt Engineering

Teach students to communicate effectively with AI to get accurate, useful results.

- Anatomy of a good prompt: context, instruction, format, constraints
- Common prompting mistakes and how to fix them
- Techniques: role prompting, step-by-step reasoning, giving examples
- Verifying and fact-checking AI output — teaching healthy skepticism
- Hands-on exercise: students write and refine prompts live

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Introduction to AI Agents

Move students from "using AI" to "directing AI."

- What is an AI agent? How it differs from a simple chatbot
- Real-world examples: study assistants, tutors, research agents
- Core building blocks: instructions, knowledge/context, memory, tools
- Planning an agent: identifying a real problem before building

Program Timeline (continued)

From building working agents to responsible, real-world application.

5

Building Subject-Specific Study Agents

Hands-on build session — students create their own working AI study tool.

- Step-by-step: building a doubt-clearing agent (e.g. a Physics tutor agent)
- Feeding subject knowledge/context into an agent
- Designing agents that generate notes, summaries, or practice questions
- Customizing agents for Math, Physics, Chemistry, Biology, Languages, etc.
- Testing, refining, and troubleshooting the agent's responses

6

Vibe Coding — Building Your Own AI-Powered Tools

Introduce lightweight, AI-assisted coding so students can build simple tools without prior coding experience.

- What is vibe coding? Coding with AI as a collaborator
- Building a simple interface for their study agent
- Connecting their agent to a usable platform/tool
- Live build: students create a simple personal "AI study platform"
- Showcase: students present what they've built

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Responsible AI Use & Future Pathways

Leave students with a responsible mindset and awareness of where these skills can lead.

- Recap: using AI as a tool for growth, not a shortcut
- Data privacy and safety basics when using AI tools
- Career and study pathways involving AI
- Open Q&A, discussion, and feedback collection

Learning Outcomes

By the end of this program, students will be able to:

- 1 Explain what AI is and how it works at a basic conceptual level
- 2 Write effective prompts to get accurate, useful results from AI tools
- 3 Critically evaluate AI-generated output rather than accepting it blindly
- 4 Build a working, subject-specific AI study agent for their own use
- 5 Use vibe coding to create a simple personal AI-powered study tool
- 6 Apply AI responsibly and ethically in an academic setting

Suggested Session Structure

Flexible — can be compressed into fewer, longer sessions depending on available time.

Session	Focus
Session 1	Module 1 — Understanding AI + Module 2 — AI in Everyday Studies
Session 2	Module 3 — Prompt Engineering
Session 3	Module 4 — Introduction to AI Agents
Session 4	Module 5 — Building Subject-Specific Study Agents
Session 5	Module 6 — Vibe Coding
Session 6	Module 7 — Responsible AI Use + Final Showcase

Recommended: 6 sessions of 60–90 minutes each, or condensed into a 2-day intensive workshop.